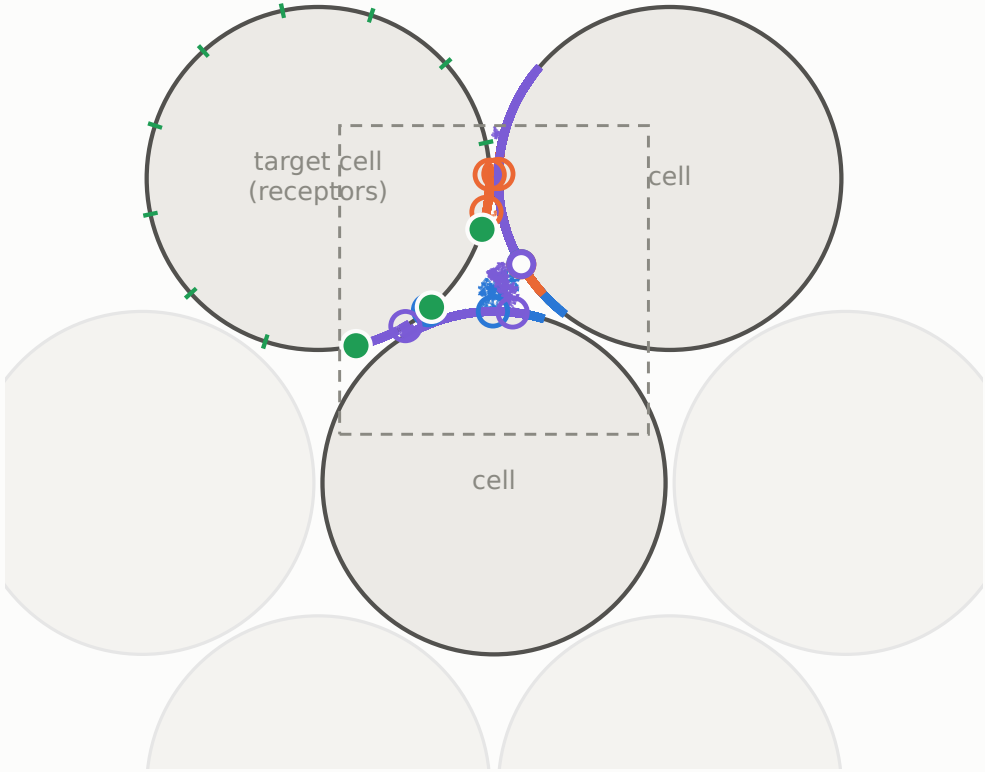
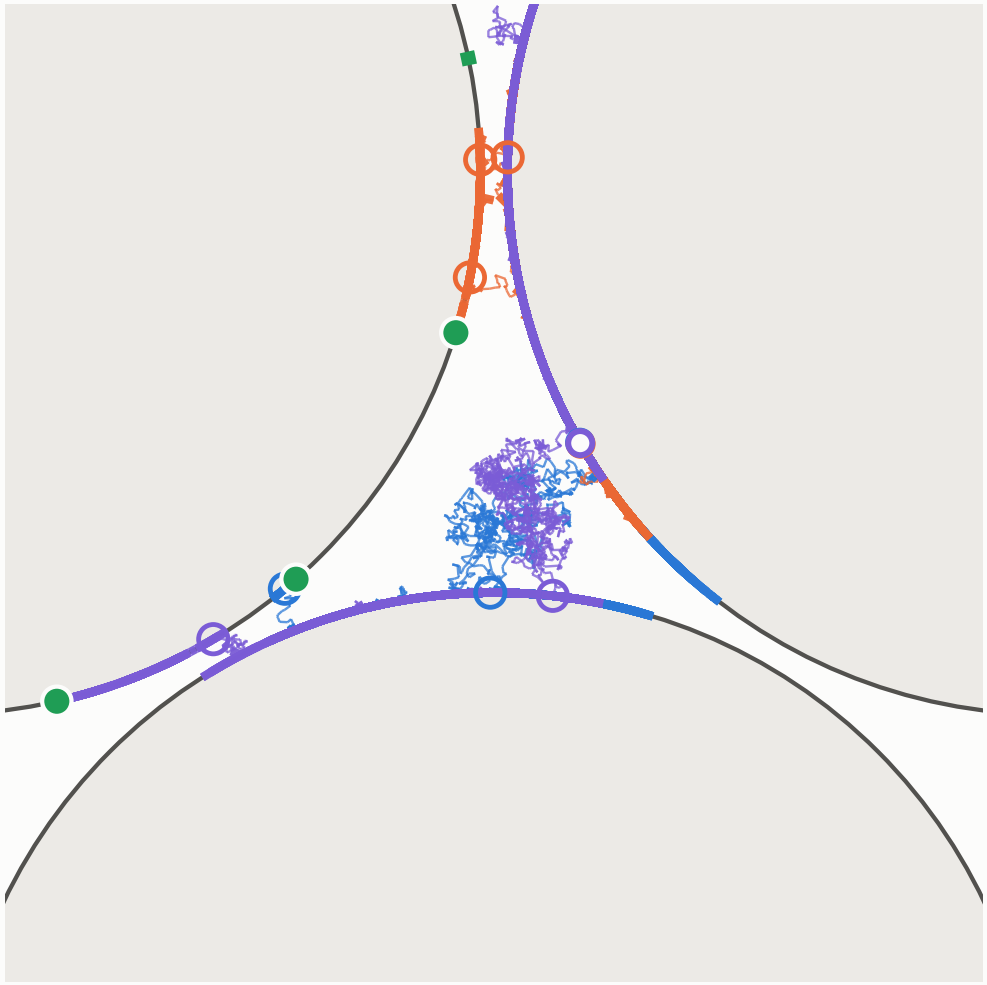
Where the search happens: sliding along a cell, then hopping the slit to the next

Seven cells, packed: the molecule never leaves the gaps



The three-cell junction, 18 um across



sliding on a membrane (1D) — free in the gap (2D) — lands on a different cell — finds a receptor

R = 10 um cells, 0.5 um gap, D = 0.5 um^2/s free and bound, mean dwell 1 s, about 7% of the time free. 3 molecules, all started bound at the same point on the lower cell of the junction; receptors on one cell only. Found a receptor 44 s after 2 hops, 76 s after 3 hops, 91 s after 2 hops. Over 150 molecules the median find is 109 s (quartiles 60 and 265 s) with 2 hops, and 29% have not found one by 1500 s; these three were picked from the first 90 seeds as paths that both hop and find quickly enough to stay legible. Cartoon only: 2D, and no statistics - the measured comparison is in the report's capture curves.